

Denoising ELVA Dataset

Final Project Report

Sölvi Santos

Email: sos106@hi.is

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1 Introduction

Capturing high-speed video presents unique challenges for computer vision systems. At 250–400 fps with multi-megapixel sensors, extremely short exposure times and limited lighting introduce strong sensor noise and occasional motion blur. For applications such as swing analysis, trajectory estimation, and markerless 3D reconstruction, frame quality directly affects downstream tasks. Effective denoising is therefore a critical preprocessing step for ELVA’s motion-capture pipeline.

This project evaluates two modern deep video denoising architectures: **FastDVDnet** [5], a lightweight flow-free model designed for real-time inference, and **SwinIR** [2], a transformer-based restoration network emphasizing high perceptual quality. These models represent complementary design philosophies: efficiency and temporal stability versus attention-driven detail preservation. Only FastDVDnet is fine-tuned in this work, while SwinIR is included as a zero-shot baseline.

Both are tested on ELVA’s multi-view high-speed golf recordings, which contain gain-amplified noise, clipping, and compression artifacts. This dataset provides a demanding and realistic environment that differs substantially from standard synthetic benchmarks.

The goal is to determine whether zero-shot pretrained models or a self-supervised fine-tuned variant of FastDVDnet can meet ELVA’s accuracy and latency requirements. Our evaluation includes pretrained inference, ELVA-specific fine-tuning, and a comparative analysis across NIQE, variance reduction, runtime, and compression-based metrics. The remaining sections cover the dataset and methodology (Section 2), experimental results (Section 3), system-level discussion (Section 4), and conclusions with recommendations (Section 5).

2 Methods

2.1 Dataset

2.1.1 Elva Golf Dataset

The Elva Golf dataset contains roughly 50,000 multi-view recordings captured with hardware-synchronized high-speed cameras. The main capture setup uses six RGB cameras operating at 5-megapixel resolution at 250 fps or higher. In addition to this configuration, there is a smaller dataset collected at ELVA headquarters using twelve-camera system with 3-megapixel sensors. These cameras run at the same nominal frame rate, and in some cases, reach 400+ FPS.

The primary sources of noise are a direct consequence of the imaging conditions. Exposure times are extremely short (around 700 microseconds), and lighting can be limited, which forces the system

to apply substantial analog gain. This combination produces strong sensor noise and clipping-related artifacts. In addition, the dataset includes compression artifacts introduced during video encoding. Although the raw uncompressed data is available, all experiments in this work operate on the compressed RGB recordings, which reflect the conditions encountered in the deployed systems.

2.1.2 Validation Window Sampling

Evaluating on the full dataset is impractical, so a fixed validation subset is defined using "windows" each specifying a swing ID, camera ID, and a temporal center frame. These windows are stored in manifest files and provide a reproducible evaluation basis for all models.

A total of 40 validation windows were sampled across different swings and camera viewpoints to ensure coverage of varied capture conditions. For each window, all quantitative metrics (NIQE, standard-deviation reduction, runtime, and compression-based measurements) are computed on the corresponding full-resolution frames. This design guarantees that pretrained FastDVDnet, fine-tuned FastDVDnet, and the SwinIR pilot are evaluated on exactly the same spatiotemporal content. Visual comparisons such as triptychs and zoom-ins are also derived from these windows. The pilots were evaluated on another larger set before training to get a feel of how the models perform.

2.2 Preprocessing Pipeline

2.2.1 Frame Handling

All experiments operate directly on the RGB video frames extracted from the ELVA recordings. For each entry in the validation manifest, the corresponding full-resolution frame is loaded without any resizing, downsampling, gamma correction, or color-space adjustments. The models therefore receive the data in their native uint8 RGB format, maintaining the sensor noise attributes and compression artifacts inherent in the original footage.

Spatial cropping is avoided during evaluation, as altering the frame area would change the local noise statistics and bias metrics such as NIQUE or standard deviation. Crops are exclusively used for visualization and qualitative assessment but are never involved in quantitative evaluation. By doing this we ensure that all models are evaluated under realistic settings that correspond to their intended implementation within the ELVA processing pipeline.

2.2.2 Metrics and Normalization

The evaluation uses four metrics aligned with ELVA’s requirements: (i) NIQUE for perceptual naturalness, (ii) standard-deviation reduction for noise suppression, (iii) per-frame runtime for deployability, and (iv) compression bitrate as an indirect perceptual measure. We report deltas relative to the noisy frame. $\Delta\text{NIQE} = \text{NIQE}_{\text{denoised}} - \text{NIQE}_{\text{noisy}}$ (negative values mean perceptual degradation), and $\Delta\sigma = \sigma_{\text{noisy}} - \sigma_{\text{denoised}}$ (larger values indicate stronger variance reduction).

The evaluation pipeline calculates all performance metrics straight from the model outputs, maintaining them in uint8 RGB format for consistency with the input data. Before calculating NIQUE [3], each frame is converted to grayscale using standard luminance coefficients and thereafter transmitted unchanged to the NIQUE estimator. The decrease in standard-deviation is derived from the RGB data by comparing the pixel-level variance of the denoised output with that of the noisy input.

Normalization is applied only immediately before inference and follows the requirements of each model. FastDVDnet expects inputs in the $[0, 1]$ range, so frames are scaled by $1/255$ and rescaled after

inference. SwinIR operates on similarly normalized tensors but does not require additional preprocessing. No spatial tiling is applied during inference, and all models process each frame in a single forward pass. The only use of spatial patches occurs during FastDVDnet fine-tuning, where 256×256 crops are sampled to fit the temporal window into GPU memory. Evaluation always uses full-frame inference.

2.3 Models

FastDVDnet (2.48 M paramters) is a lightweight, flow-free video denoising network that processes five-frame temporal windows and aggregates information using convolutional subnetworks. Its architecture is designed for real-time use and does not rely on optical-flow estimation. In this work it is used both in its pretrained zero-shot form and, separately, as a model fine-tuned on ELVA data (details in Section 2.4).

SwinIR (11.50 M paramters) is a transformer-based restoration model built from hierarchical Swin Transformer blocks that capture long-range spatial dependencies. It is included solely as a high-capacity zero-shot baseline to assess how a modern attention-based architecture behaves on ELVA footage without domain adaptation.

2.4 Training Procedure (FastDVDnet POST)

Fine-tuning of FastDVDnet was performed using a self-supervised Noise2Void-style objective [1] adapted to the characteristics of the ELVA dataset. Each training sample consists of a temporal window of five consecutive frames from a single camera stream, from which a 256×256 spatial crop is extracted. The center frame serves as the prediction target, while a randomly masked version of the same frame is used as input. Masking is applied to a small fraction of pixels using a blind-spot mechanism that replaces each selected pixel with the average of its local neighborhood. The masking probability increases gradually over the course of training. This formulation encourages the model to infer the underlying clean structure from context without requiring clean ground-truth data.

Optimization uses the AdamW optimizer with a learning rate of 2×10^{-5} and weight decay of 10^{-4} . The newest training run was conducted for 1,000 steps, matching the length of the cosine warmup schedule. So, the warmup and decay operate over the full run: the learning rate ramps linearly during the initial phase and then decays following a cosine profile. In hindsight, a shorter warmup or a longer overall schedule may have been more appropriate, since the effective learning rate remained low for a large portion of training. Mixed-precision training (AMP) is enabled throughout, and gradient accumulation over 16 iterations is used to achieve an effective batch size of one due to memory constraints at the native resolution. An exponential moving average (EMA) of the model parameters with decay 0.9995 is maintained and used during validation. Validation is performed every 50 steps on 32 masked windows and the final EMA-corrected checkpoint is used for all evaluation

2.5 Evaluation

All models are evaluated on the fixed set of 40 validation windows described in Section 2.1.2. For each window and model, the denoised center frame is compared against the original noisy frame using a small set of frame-level metrics. NIQE is computed on a luminance-converted version of each frame, while standard-deviation reduction is measured on the RGB data to quantify how much apparent noise variance is removed. Runtime is recorded as the wall-clock time required to denoise a single center frame, aggregated into mean and percentile statistics across all windows. These metrics are evaluated

on full-resolution frames to preserve the native noise characteristics and avoid artifacts associated with downsampling or cropping.

In addition to these image-level metrics, a compression-based score is computed by re-encoding short video clips composed of noisy and denoised frames using ELVA’s libav-based H.264 pipeline. All clips are encoded with identical settings (CRF=18, keyframe interval 48), ensuring comparability across models. Compression bitrate is used as a proxy for perceptual quality, since smoother and less noisy videos typically produce smaller encoded file sizes at a fixed quality setting. Finally, qualitative assessment is carried out through triptych visualizations showing noisy input, pretrained FastDVDnet, fine-tuned FastDVDnet, and SwinIR outputs side by side with zoomed-in crops.

3 Results

3.1 Pilot Comparison FastDVDnet vs SwinIR

Both pretrained models were initially evaluated on the same 357 full-resolution windows. Qualitatively, both FastDVDnet and SwinIR suppress high-frequency sensor noise, but both also introduce characteristic oversmoothing and slightly exaggerated edges (Figure 3). The standard-deviation histogram shows that SwinIR provides more stable noise reduction, consistently above 0.001, whereas FastDVDnet varies more widely. The NIQE-delta distributions indicate that both models reduce perceptual naturalness, with FastDVDnet shifted further into negative values and SwinIR closer to zero. The dominant practical difference is speed: FastDVDnet achieves roughly 0.4 s per full-resolution frame, while SwinIR requires about 20 s (Figure 2). This 50x difference, along with the potential for domain adaptation, motivates focusing the remaining experiments on FastDVDnet.

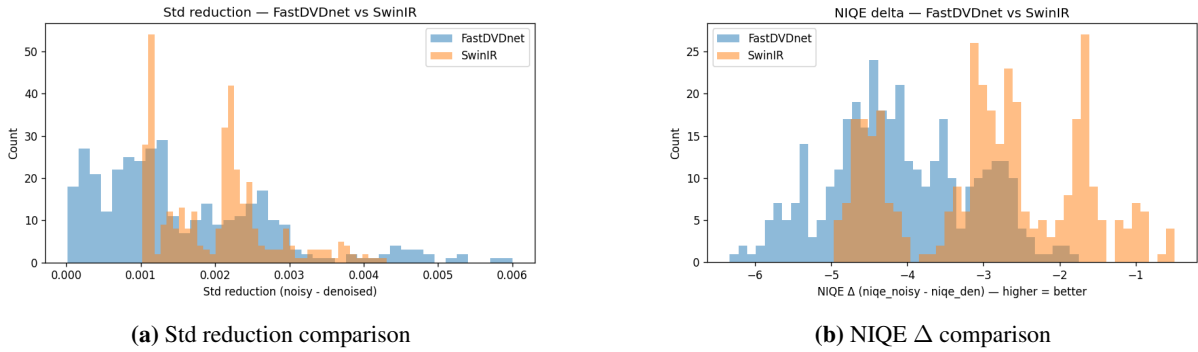


Figure 1: Distribution of std-reduction and NIQE-delta across 357 windows.

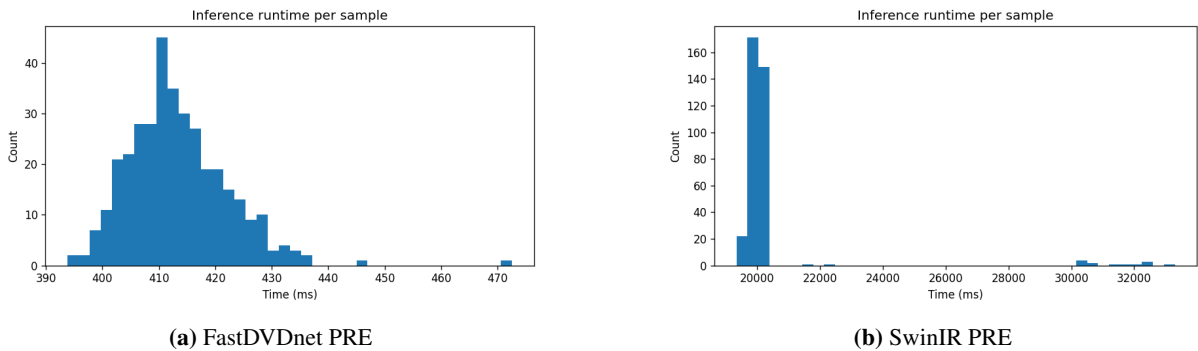


Figure 2: Inference runtime distribution for both pretrained models.

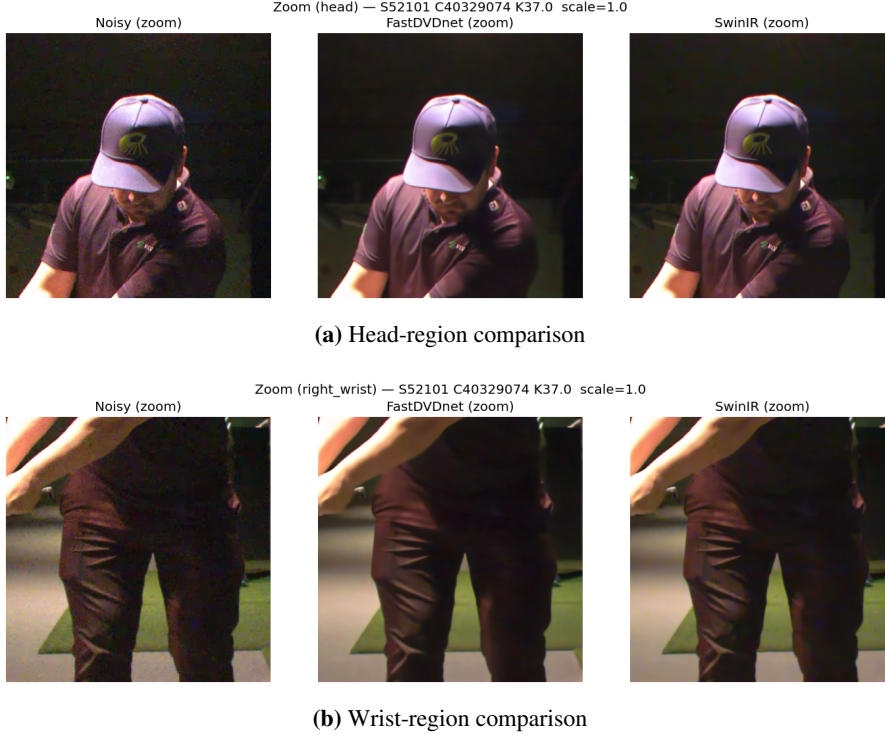


Figure 3: Zoomed triptychs comparing noisy input, FastDVDnet PRE, and SwinIR PRE.

Model	$\Delta\text{NIQE} \uparrow$	Std. reduction $\uparrow$	Runtime (s) $\downarrow$
FastDVDnet-PRE	-4.09	0.00155	0.41
SwinIR	-2.93	0.00202	20.41

Table 1: Pilot metrics for FastDVDnet PRE vs. SwinIR PRE.

3.2 Comparison of Pilots vs. FastDVDnet Fine-Tuned Model

Across the validation set, all three models reduce sensor noise but exhibit distinct perceptual behaviors. As illustrated in the wrist-level crops in Figure 4, FastDVDnet-PRE produces the strongest visible smoothing, giving it the characteristic “plasticky” appearance seen in Figures 3 and 4. Although POST slightly increases the measured standard-deviation reduction (Table 2), it does not smooth as aggressively in practice: its outputs retain more low-frequency residual noise, which makes the images appear less uniform than PRE. This is reflected in the more negative ΔNIQE values for POST, indicating reduced perceptual naturalness.

SwinIR preserves the most high-frequency detail and achieves the best ΔNIQE among the three models, but the inference cost is prohibitive: runtimes average roughly 20 s per frame (Figure 5). In contrast, both FastDVDnet variants operate near 0.4 s per frame, making them the only viable options for ELVA’s real-time pipeline. Compression analysis (Appendix A) shows that all three models substantially reduce bitrate compared to the noisy input, with SwinIR typically achieving the lowest bits-per-pixel-per-frame. FastDVDnet-POST is slightly more compressible than PRE on some sequences but worse on others, indicating that its reduced oversmoothing does not translate into a consistent bitrate advantage.

Overall, Table 2 and Figure 5 highlight a clear trade-off: PRE produces the cleanest-looking but most oversmoothed outputs, POST reduces variance slightly more numerically but looks less natural, and SwinIR delivers the best perceptual quality at an impractical computational cost.

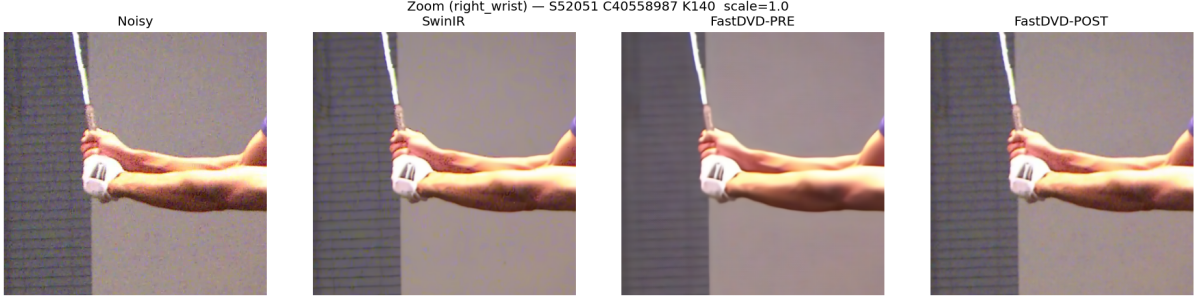


Figure 4: Wrist-region comparison. FastDVDnet-PRE and POST exhibit smoothing; SwinIR preserves sharper detail.

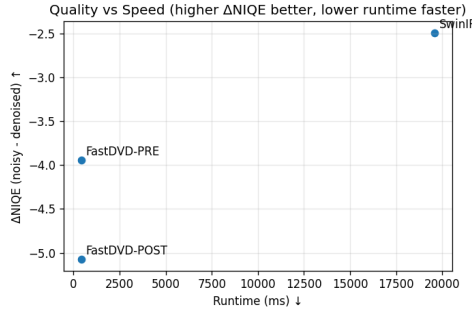


Figure 5: Quality-speed tradeoff across all models. Higher ΔNIQE indicates more natural output; lower runtime indicates faster inference.

Model	$\Delta\text{NIQE} \uparrow$	Std. reduction $\uparrow$	Runtime (s) $\downarrow$
FastDVDnet-PRE	-3.94	0.00173	0.44
FastDVDnet-POST	-5.07	0.00198	0.41
SwinIR	-2.49	0.00221	19.58

Table 2: Final comparison across all models (seconds per full-resolution frame).

4 Discussion

4.1 Why FastDVDnet Pretrained Works a bit too Well

The pretrained FastDVDnet model was optimized on synthetic Gaussian noise with clean targets, which does not fully match ELVA’s gain-amplified, clipped and compressed footage (Section 2.1). On ELVA data, the network operates on five-frame windows and implicitly averages over small motions and textures, which suppresses random sensor noise but also removes fine-grained detail that is relevant for pose estimation. The pilot and final metrics (Tables 1 and 2) reflect this behavior: FastDVDnet-PRE delivers consistent standard-deviation reduction but pushes the mean ΔNIQE far into the negative range. Visually, this appears as overly smooth skin and clothing in Figures 3, 4, and 7, where much of the motion is preserved but the images look "plasticky" rather than natural.

4.2 Why FastDVDnet Self-Supervised Fine-Tune Failed in this Instance

The self-supervised fine-tuning was intended to adapt FastDVDnet to ELVA’s noise characteristics without clean ground truth (Section 2.4). In practice, the POST model only slightly increases standard-

deviation reduction while further degrading ΔNIQE (Table 2), suggesting that it has partly internalized structured noise and compression artifacts instead of removing them. This is consistent with known limitations of Noise2Void-style [1] objectives on correlated or non-stationary noise: when artifacts dominate local statistics, the model’s conditional expectation can converge toward those artifacts. In our setting, sensor noise is correlated across frames and compression introduces blockwise patterns, so the blind-spot formulation was not strong enough to cleanly separate noise from texture. As a result, FastDVDnet-POST smooths marginally less than PRE but can reintroduce low-frequency noise patterns, visible in some crops in Figure 4 and Figure 6, where residual structured noise remains despite the apparent variance reduction. The training schedule itself was also shorter than intended: running beyond 1,000 steps was not feasible due to I/O bottlenecks and project-deadline constraints, which further limited the model’s ability to adapt cleanly to ELVA’s noise characteristics.

4.3 SwinIR Trade-Offs

SwinIR provides the most detailed and natural-looking reconstructions among the evaluated models. It achieves the least negative ΔNIQE and the strongest mean standard-deviation reduction (Table 2), and visually preserves high-frequency structure better than either FastDVDnet variant (Figures 3 and 4). However, this comes at a computational cost that is incompatible with ELVA’s real-time requirements: runtimes are around 20 s per full-resolution frame, more than an order of magnitude slower than FastDVDnet (Figures 2 and 5). Part of this disparity can be attributed to model capacity: SwinIR has approximately 11.5 M parameters compared to FastDVDnet’s 2.48 M, contributing both to its stronger detail preservation and its substantially higher inference cost. SwinIR is therefore best viewed as a high-fidelity reference or an option for offline analysis of selected recordings, rather than a realistic candidate for deployment in the current pipeline.

4.4 I/O Constraints

A major practical limitation in this project was the cost of data loading. Each FastDVDnet training step requires only a temporal window of $T = 5$ frames, but the underlying compressed recordings contain hundreds of full-resolution frames. Because the videos are stored as YUV streams, accessing any subset of frames requires decoding from the beginning of the file, so every training sample effectively triggers a full-video decode at 2464×2064 resolution. Logs from the training runs show typical decode times of 1.2–1.6 s per video, while a FastDVDnet forward pass on a full-resolution frame takes roughly 0.4 s (Table 2). In other words, I/O is 3–4 $\times$ slower than the model itself and dominates the training loop, preventing the GPU from being fully utilized and limiting the number and diversity of fine-tuning iterations. This constraint likely contributed to the modest and somewhat unstable gains observed for FastDVDnet-POST.

4.5 Future Work

Several directions remain open for improving denoising performance and addressing the limitations encountered in this project. The main constraint is the absence of clean ground truth: the self-supervised objective used for FastDVDnet fine-tuning (Section 2.4) struggles with the structured gain-and-clip noise in ELVA recordings. A more reliable strategy would be to construct a dedicated training set under controlled noise conditions. Using the same cameras with systematically increased gain, reduced lighting, or varied exposure would produce paired "extreme-noise" / "normal-noise" footage, enabling supervised

or paired self-supervised training and giving the model clearer cues for separating texture from sensor noise.

A second direction is to evaluate alternative denoisers. Architectures such as EDVR, BasicVSR, or modern diffusion-based video restorers may offer stronger temporal consistency than FastDVDnet or SwinIR. EDVR is particularly promising for motion- and compression-heavy data but was not explored here due to package conflicts and the engineering effort required to run it at ELVA resolutions. A more unified data pipeline would make it easier to assess whether the performance ceiling observed in Section 3 is architectural or dataset-driven.

Infrastructure improvements could also expand what is feasible. As discussed in Section 4.4, data loading dominates training time: decoding a single video (1–1.6 s) is several times slower than a FastDVDnet forward pass. A cached dataset of pre-decoded RGB frames or pre-extracted windows would enable longer and more diverse fine-tuning schedules. Access to RAW sensor data could further help by removing compression artifacts and supporting more physically informed noise models.

Finally, there is long-term potential in leveraging ELVA’s multi-view capture directly for denoising. With 6 or 12 synchronized cameras observing the same scene, cross-view denoising by combining geometry-aware alignment with temporal modeling could exploit redundancy across viewpoints to suppress noise while preserving true structure, moving beyond single-view restoration.

5 Conclusion

In this project we evaluated two state-of-the-art video denoising architectures, FastDVDnet and SwinIR, on ELVA’s multi-view high-speed golf recordings. The goal was to assess whether off-the-shelf models, and a self-supervised fine-tuned variant of FastDVDnet, can improve visual quality and temporal consistency under realistic deployment constraints. The dataset is characterized by short exposures, high analog gain, clipping, and compression artifacts, making it substantially more challenging than the synthetic benchmarks on which these models were originally trained.

The pilot comparison showed that both pretrained models suppress sensor noise effectively but at a noticeable perceptual cost. FastDVDnet-PRE delivers stable standard-deviation reduction at around 0.4 s per full-resolution frame, but produces overly smooth, "plasticky" outputs and strongly negative Δ NIQE. SwinIR achieves higher perceptual quality and stronger average noise suppression, yet is roughly 50 $\times$ slower (about 20 s per frame), which makes it infeasible for integration into ELVA’s real-time processing pipeline.

Self-supervised fine-tuning of FastDVDnet using a Noise2Void-style objective yielded mixed results. The POST model slightly increased standard-deviation reduction relative to the pretrained baseline but further degraded Δ NIQE, indicating that it partially relearned structured noise instead of removing it. Combined with severe I/O bottlenecks during training, where video decoding dominates step time, this limited the extent to which domain adaptation could meaningfully improve perceptual quality.

Overall, the experiments indicate that FastDVDnet is more compatible with ELVA’s latency requirements than SwinIR, but its current behavior reflects a clear trade-off between temporal stability and perceptual realism. Achieving substantial gains will require more than minor fine-tuning: better curated training data, improved data-loading infrastructure, and possibly new architectures that exploit ELVA’s multi-view geometry. The work in this report provides a baseline and a set of concrete measurements that can guide these future developments toward a practical, production-ready denoising pipeline.

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A Appendix A: Compression

All compression-based measurements in Section 3 were computed by re-encoding short video clips composed of noisy and denoised center frames. Compression was performed using ELVA’s internal libav pipeline through the `compress_videos` utility. Each clip was encoded using

- Constant-rate-factor: CRF = 18
- Keyframe interval: 48
- Encoder: libav / H.264

To verify the codec configuration, one of the generated MP4 files was inspected with `ffprobe`. The encoder metadata confirmed:

```

codec_name      = h264
profile         = Main
pix_fmt        = yuv420p
r_frame_rate   = 25/1

```

Bits-per-pixel-per-frame (bpppf). To make compression results comparable across resolutions and clip lengths, all reported bitrates are expressed as bits-per-pixel-per-frame (bpppf). Given a compressed video with file size B bytes, spatial resolution $H \times W$, and n frames, bpppf is defined as

$$\text{bpppf} = \frac{8B}{H \cdot W \cdot n}$$

This normalization removes the influence of frame count and resolution, so lower bpppf directly indicates better compressibility at a fixed encoder configuration. Since all variants (noisy, PRE, POST, SwinIR) are encoded with identical settings, relative differences in bpppf mainly reflect differences in residual noise and temporal smoothness rather than codec parameters.

Swing	Camera	frames	H	W	bpppf Noisy	bpppf PRE	bpppf POST	bpppf SwinIR	Δ bytes POST-PRE	Δ bytes Swin-PRE	Δ bytes Swin-POST
52002	40436941	2	2064	2464	0.700	0.291	0.429	0.265	175514	-32553	-208067
52004	40436942	2	2064	2464	0.680	0.290	0.426	0.268	172381	-27901	-200282
52006	40436941	2	2064	2464	0.678	0.297	0.425	0.254	162771	-54106	-216877
52008	40436943	2	2064	2464	0.529	0.492	0.427	0.148	-82529	-437905	-355376
52009	40355842	2	2064	2464	0.568	0.288	0.383	0.179	120961	-138244	-259205
52012	40355842	2	2064	2464	0.820	0.256	0.464	0.303	264603	59526	-205077
52014	40436942	2	2064	2464	0.691	0.250	0.409	0.265	202896	19599	-183297
52016	40436941	2	2064	2464	0.662	0.299	0.425	0.257	160820	-52834	-213654

Table 3: Compression comparison (Top-K cameras by aligned centers). Lower bpppf and negative Δ bytes indicate better compressibility.

B Appendix B: Reproducibility

B.1 Software Environment

All experiments were conducted in Python using PyTorch [4] with CUDA acceleration. The environment configuration was:

- Python 3.10

- PyTorch 2.x (CUDA 12.x build)
- NVIDIA GPU hardware
- Dependencies for image I/O, NIQE, and compression from the ELVA internal `gav` package

B.2 Training Configuration

For FastDVDnet fine-tuning (POST), the following settings were used:

- Optimizer: AdamW (2×10^{-5} , weight decay 10^{-4})
- Schedule: Cosine warmup over 1000 steps
- AMP-level mixed precision
- Gradient accumulation: 16 steps
- Effective batch size: 1 window (5 frames, 256×256 crops)
- EMA decay: 0.9995
- Validation every 50 steps on 32 masked windows

C Appendix C: Additional Figures

This section contains supplementary qualitative comparisons not included in the main text. These figures illustrate typical behaviors of the pretrained and fine-tuned FastDVDnet models on additional validation windows.

C.1 Noisy vs. FastDVDnet-POST (After Training)



Figure 6: Additional comparison showing the effect of FastDVDnet self-supervised fine-tuning on a representative validation frame.

C.2 Noisy vs. FastDVDnet-PRE vs. FastDVDnet-POST

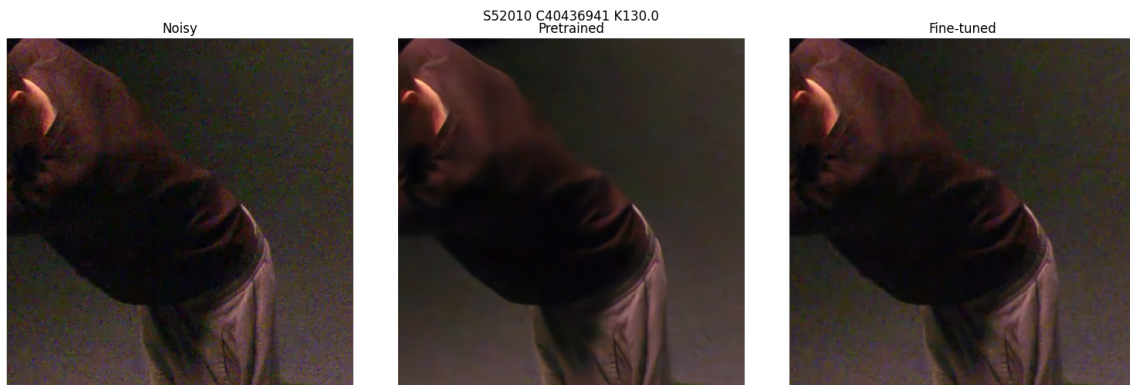


Figure 7: Supplementary triptych comparing the noisy input, FastDVDnet pretrained output, and the fine-tuned FastDVDnet-POST output, highlighting the “plasticky” appearance of the pretrained model.

D Appendix D: Summary of Evaluated Models

This appendix reports aggregated performance metrics for all evaluated models. For each method, the table summarizes the mean Δ NIQE, standard-deviation reduction, average inference time per full-resolution frame, and the 90th-percentile runtime across the 40 validation windows.

Table 4: Aggregate metrics per model (higher Δ NIQE/Std reduction is better, lower runtime is better).

Model	Params (M)	N	Δ NIQE ($\uparrow$)	Std red. ($\uparrow$)	Runtime ms ($\downarrow$)	p90 Runtime
FastDVD-POST	2.48	40.000	-5.075	0.002	415.000	424.900
FastDVD-PRE	2.48	40.000	-3.942	0.002	440.400	426.400
SwinIR	11.50	40.000	-2.487	0.002	19580.100	19511.600